

Machine Learning: Intro & Trends

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OUTLINE

- What is machine learning?
- What are the techniques
- Examples of new impressive developments
 - > ImageNet
 - > AlphaGo
 - > Generative AI
 - > Medical domain (if time)
- Conclusions

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AI DISCIPLINES

Turing test capabilities	Definitions
Natural language processing	to enable it to communicate successfully in English.
Knowledge representation	to store what it knows or hears.
Automated reasoning	to use the stored information to answer questions and to draw.
Machine learning	to adapt to new circumstances and to detect and extrapolate patterns.
Computer vision	to perceive objects.
Robotics	to manipulate objects and move about.

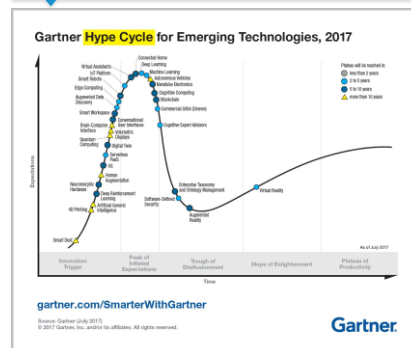
Source: Russel and Norvig (2009, p. 2)

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THE HYPE CYCLE (GARTNER)



Gartner.



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MACHINE LEARNING

- How do we define it?
 - > “Machine learning is to automatically identify patterns from data”
 - > “A computer program is said to learn from experience E with respect to some class of tasks T and performance P, if its performance at tasks in T improves with E.” (Mitchell)
- Why did it become so popular?
 - > Increasing amount of data available
 - > More computational power
 - > Better algorithms

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MACHINE LEARNING TASKS

- What are machine learning tasks?
 - > **Supervised learning** is the machine learning task of inferring a function from a set of labeled training data
 - > In **unsupervised learning**, there is no target measure (or label), and the goal is to describe the associations and patterns among the attributes
 - > **Reinforcement learning** tries to find optimal actions in a given situation so as to maximize a numerical reward that does not immediately come with the action but later in time.

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WHAT STEPS ARE NEEDED?

Table 1 Side-by-side comparison of the major existing KDDM models

Model	Fayyad <i>et al.</i>	Cabena <i>et al.</i>	Anand & Buchner	CRISP-DM	Les <i>et al.</i>	Generic model
Area	Academic	Industrial	Academic	Industrial	Academic	N/A
No. of steps	9	5	6	6	6	6
Refs	(Fayyad <i>et al.</i> , 1996)	(Cabena <i>et al.</i> , 1998)	(Anand & Buchner, 1998)	(Shearer, 2000)	(Les <i>et al.</i> , 2000)	N/A
Steps	1: Developing and Understanding of the Application Domain 2: Creating a Target Data Set 3: Data Cleaning and Preprocessing 4: Data Reduction and Projection 5: Choosing the DM Task 6: Choosing the DM Algorithm 7: DM 8: Interpreting Mixed Patterns 9: Consolidating Discovered Knowledge	1: Business Objectives Determination 2: Data Preparation 3: DM 4: Domain Knowledge Elicitation 5: Animation of Knowledge	1: Human Resource Identification 2: Data Prospecting 3: Domain Knowledge Elicitation 4: Methodology Identification 5: Data Preprocessing 6: Deployment	1: Business Understanding 2: Data Understanding 3: Data Preparation 4: Modeling 5: Evaluation 6: Deployment	1: Understanding the problem Domain 2: Understanding the Data 3: Preparation of the Data 4: DM 5: Evaluation 6: Using the Discovered Knowledge	1: Application Domain Understanding 2: Data Understanding 3: Data Preparation and Identification of DM Technology 4: DM 5: Evaluation 6: Knowledge Consolidation and Deployment

L. A. KROGAN AND P. MOUTIER

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WHICH STEP TAKES MOST TIME?

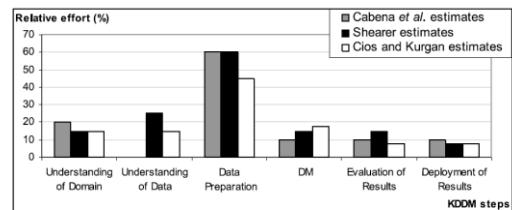


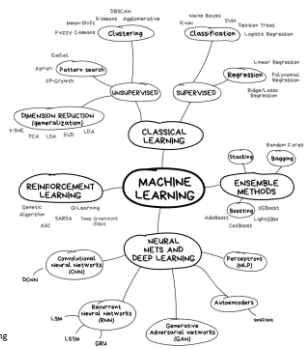
Figure 1 Relative effort spent on specific steps in the KDDM process

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MACHINE LEARNING ALGORITHMS



Source: vas3k's blog

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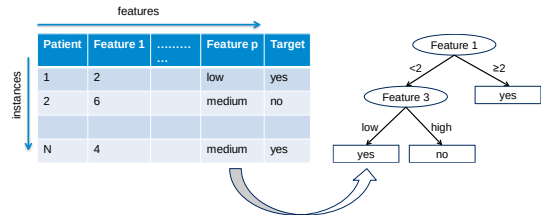


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SUPERVISED LEARNING

- Supervised learning approaches learn using a labeled dataset:

> "Regular" machine learning algorithms:

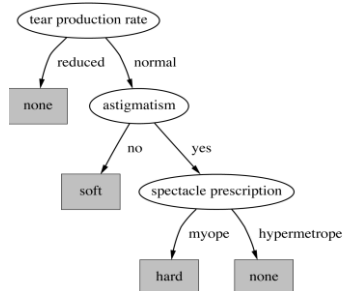


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ALGORITHM EXAMPLE 1: DECISION TREE

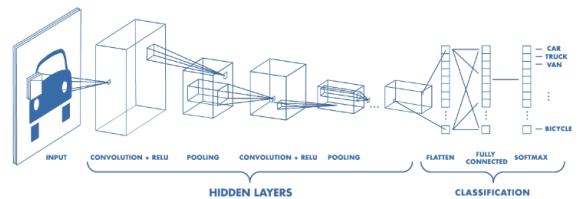


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ALGORITHM EXAMPLE 2: NEURAL NETWORK



Source: towardsdatascience.com

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EVALUATION

- Want to have models that predict well for new examples
- We train on a training set and apply our algorithms to an independent test set to assess generalizability
- This will show us how good it will work on new examples

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DO WE KNOW WHAT IT LEARNS?

- Important to realize that for some models we are really talking about black boxes
- Some horror from the past around this...



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LET'S JUMP INTO SOME IMPRESSIVE EXAMPLES!

- Imagenet and deep learning
- Alpha go
- Generative AI
- Examples from the medical domain

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IMAGENET

ImageNet Challenge

IMAGENET

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.



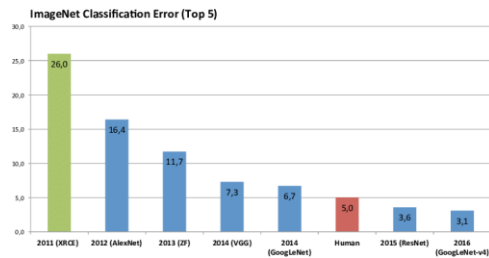
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PERFORMANCE SURPASSED THAT OF A HUMAN

- Performance over the years:

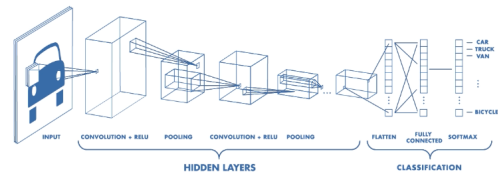


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DEEP LEARNING (1)



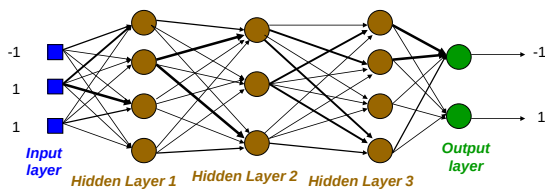
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DEEP LEARNING (2)

- Neural networks: composed of thousands of very simple computational units



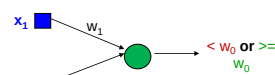
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DEEP LEARNING (3)

- Connections have a strength (also called a weight)
- Neurons multiply the weight with the value of the input



- $x_1 * w_1 + x_2 * w_2$
- Then we apply some function

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DEEP LEARNING (4)

- Challenge: how do we learn these weight?
 - > We use our network to predict what picture we are seeing
 - > If we make a mistake we "punish" the weights that contributed most to the mistake, and adjust them to avoid making the same mistake again
 - > We continue this for our entire training set
- Problem: there are soooo many weights
 - > The more weights we have, the more data we need to train
- Deep learning: reduces the number of weights through special layers

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DETECTION OF SKIN CANCER



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RECOGNIZING SCENES

- Recognizing the scene in images



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Vinyals et al., Show and Tell: A Neural Image Caption Generator



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RECOGNIZING SCENES

- Recognizing the scene in images

Approach	PASCAL (xfer)	Flickr 30k	Flickr 8k	SBU
Im2Text [24]	25	55 56	48 58 51	11
TreeTalk [18]				19
BabyTalk [16]				
Tri5Sem [11]				
m-RNN [21]				
MNLM [14]				
SOTA	25	56	58	19
NIC	59	66	63	28
Human	69	68	70	

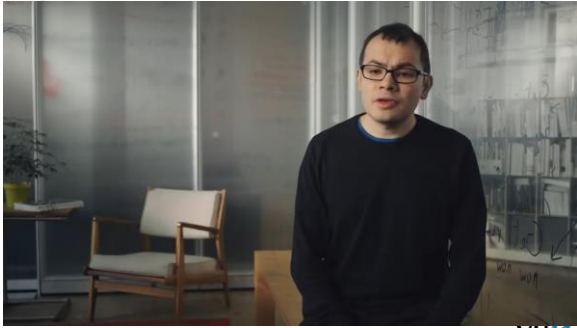
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Vinyals et al., Show and Tell: A Neural Image Caption Generator



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ALPHAGO



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HOW DOES IT WORK UNDER THE HOOD?

- It is an example of reinforcement learning (RL)
- Within reinforcement learning we have a few ingredients:

state

action

reward



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REINFORCEMENT LEARNING (1)

- We learn to select the best action in a given state such that we maximize ***all the future reward***
- In GO: we want to win the game
- Actions in a state might give us an immediate reward (e.g. we gain an advantage on the board) but might turn out bad in the long run (we lose)
 - Sequential actions
- Of course we have lots of states (we have seen this)

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REINFORCEMENT LEARNING (2)

- So how do we learn what to do when?
- We just start selecting random actions to start with (exploration) and start learning which actions are suitable in what states (i.e. give us a lot of future reward)
- We move more to exploitation once we have found a good strategy
- We do this in a simulator (needed!)
 - Makes this less applicable for cases where we cannot build such a simulator (though developments are under way)

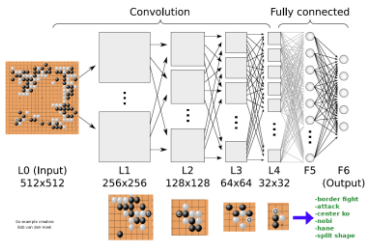
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REINFORCEMENT LEARNING (3)

- How do we deal with the huge number of states?
 - Deep reinforcement learning



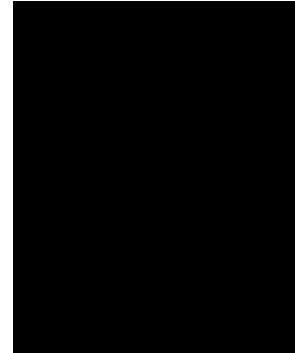
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REINFORCEMENT LEARNING (4)

- Atari breakout



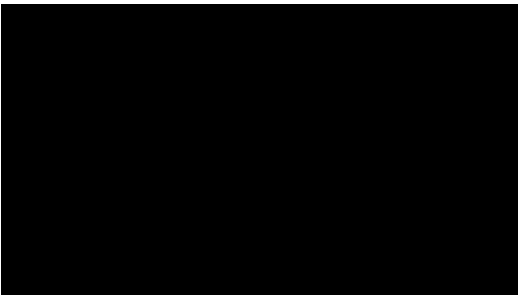
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REINFORCEMENT LEARNING (5)

- Pacman



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REINFORCEMENT LEARNING (6)

- Expected to become next big wave after deep learning
- Can also be used for personalization, e.g.
 - Netflix recommendations
 - Medical treatments
 -

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REINFORCEMENT LEARNING (7)

- We do need lots of computing power...



AlphaGO 1202 CPUs, 176 GPUs,
100+ Scientists.

Lee Se-dol 1 Human Brain,
1 Coffee.

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GENERATIVE AI

- What if we could learning to generate data
 - > We generate a picture of a car we haven't seen before
 - > We generate a picture of us looking much older (Russian app)



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GENERATIVE AI

- Generative Adversarial Networks can help us here
- They work in quite a nice way by two competing forces at work:
 - > One network tries to generate images (e.g. an old man picture based on a young picture) as good as it can
 - > The other network tries to figure out which images are real and which ones are fake
 - > So we need real data (real pictures of a person being young and being old) for this to work

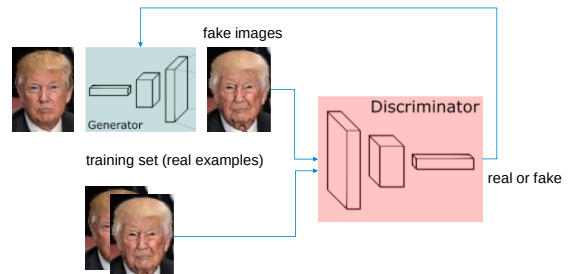
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GENERATIVE AI

- This shows the architecture



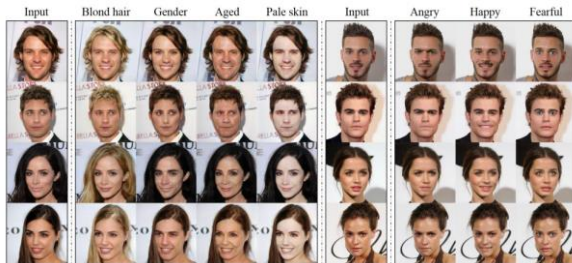
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STARGAN

- Here we see some examples of stars



Choi et al., StarGAN: Unified Generative Adversarial Networks for Multi-Domain Image-to-Image Translation



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HEALTH

- Huge increase is seen in data collected about health and wellbeing
- Health data collected in various ways:
 - > By medical staff (electronic medical records)
 - > By smart device sensors (smart phone, smart watch)
 - > By the people (prompts on your mobile phone, social media, etc.)
- On top, smart devices can easily be used as a mechanism for health interventions
 - > Providing feedback messages
 - > Providing exercises



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ELECTRONIC MEDICAL RECORDS

- Some data about Electronic Medical Records¹:
 - > Only in the US (in 2010) EMR's contained 150 exabytes of data
 - > Prospected to become yottabytes (10^{24}) in the very near future
 - > Large part of the data is *unstructured* (some say 80%), think of free text notes, medical images, etc.



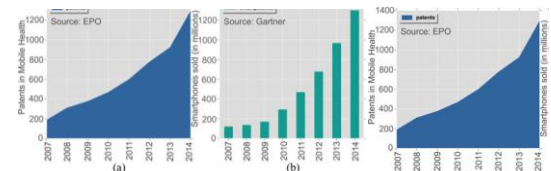
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MOBILE HEALTH

- Mobile health¹:



- Over 100,000 mobile apps in the iTunes store alone (!)²

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PROBLEMS?

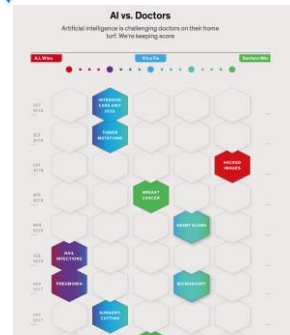
- Problem with current techniques used in the medical domain:
 - > Huge amount of data
 - > Hypothesis driven: how can we find new/unexpected things?
 - > Cannot handle more complex patterns
 - > Cannot handle unstructured data (images, etc.)
 - > Hard to personalize therapies
- Machine learning can help here

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AI IN THE MEDICAL DOMAIN

42 <https://spectrum.ieee.org/static/ai-vs-doctors>

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EXAMPLE MEDICAL DOMAIN

- Medical data is commonly specified like this:
 - Patient X, aged 55, male
 - Medication
 - 21-01-2009, A01, ...
 - 25-03-2009, C10, ...
 - 09-04-2009, C10, ...
 - ...
 - Referral
 - 19-01-2009, Cardiologist, ...
 - 21-01-2009, Surgeon, ...
 - Sensor values
 - 08-01-2009, 19:30, heart rate, 80, ...
 - ...
 - Consults
 - 21-01-2009, D01, ...
 - 26-01-2009, D01, ...
 - 20-05-2009, D75, ...
 - ...
 - Free text
 - 10-01-2009, "patient is complaining about abdominal pain", ...
 - ...

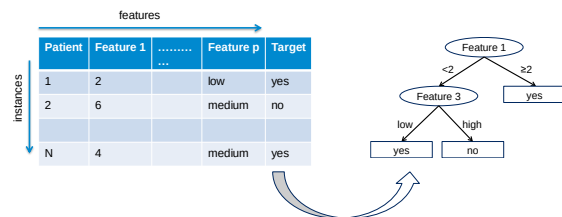
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IDENTIFICATION OF FEATURES (1)

- Machine learning approaches expect:
 - > "Regular" machine learning algorithms:



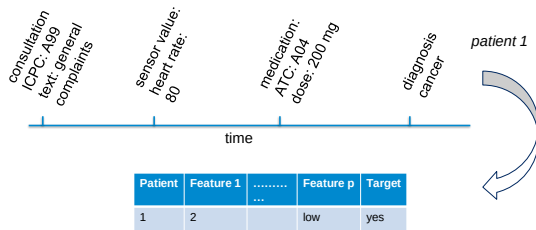
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IDENTIFICATION OF FEATURES (2)

- How can we go from our medical dataset to a dataset which is usable for regular machine learning?



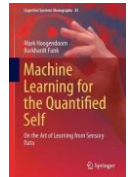
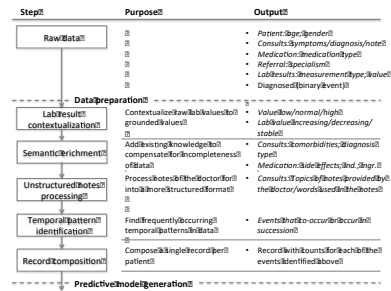
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FEATURES FROM EMR DATA

- Steps to extract features from a variety of data types for Electronic Medical Records



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CASE STUDY (1)

- Applied to colorectal cancer
- Predict colorectal cancer based on symptomatic data of the 6 months before diagnosis
- GP records of ± 380.000 Dutch citizens ($\pm 2007-2011$)
 - > 11 million consult records
 - > 23 million prescription records
 - > 23 million lab test records
 - > 4 million specialist referral records
 - > 1.2 million free text records (subset of all patients)
- Imbalanced: ± 1300 CRC patients = 0.34%

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CASE STUDY (2)

- Results (5 fold cross validation, Area Under the ROC Curve)

Algorithm Condition	CART	Logistic regression	Random Forest
Benchmark			
Age & Gender	0.831 (0.817-0.844)	0.836 (0.823-0.850)	0.834 (0.821-0.848)
Bristol-Birmingham equation	0.851 (0.838-0.864)	0.864 (0.851-0.877)	0.865 (0.873-0.897)
Regular pipeline			
LR	0.854 (0.841-0.867)	0.871 (0.859-0.884)	0.888 (0.876-0.900)
LR + SE	0.854 (0.841-0.867)	0.871 (0.859-0.884)	0.889 (0.877-0.900)
LR + Temp	0.859 (0.837-0.883)	0.883 (0.871-0.895)	0.888 (0.886-0.893)
LR + SE + Temp	0.849 (0.836-0.863)	0.879 (0.867-0.891)	0.881 (0.868-0.891)
Pipeline with only temporal patterns used for prediction			
LR + Temp	0.855 (0.842-0.868)	0.891 (0.879-0.903)	0.882 (0.870-0.894)
LR + SE + Temp	0.855 (0.842-0.868)	0.891 (0.879-0.903)	0.882 (0.871-0.895)

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CASE STUDY (3)

Predictors:

Predictor (temporal pattern)	Importance in model	CRC cases	Non-CRC cases
*Drugs for constipation	2.72	336 (26.06)	7231 (2.83)
*Iron deficiency anemia	1.88	87 (6.78)	1080 (0.43)
*Lipid modifying agents (s) *Drugs for constipation	1.77	85 (6.68)	1058 (0.43)
*Age	1.62	-	-
*Drugs for acid related disorders (s) *Drugs for constipation	1.55	99 (7.78)	2179 (0.83)
*Diabetes non-insulin dependent	0.99	161 (12.55)	8968 (3.43)
*Abdominal pain/cramps general	0.91	87 (6.78)	1687 (0.63)
*Diabetes non-insulin dependent (s) *Diabetes non-insulin dependent	0.89	132 (10.28)	7566 (2.83)
*Beta blocking agents (s) *Drugs for constipation	0.86	68 (5.33)	1111 (0.43)
*Hypertension uncomplicated (s)	0.79	168 (13.08)	9921 (3.83)
*Agents acting on the renin-angiotensin system (s) *Drugs for constipation	0.63	83 (6.43)	1208 (0.53)
*Diuretics	0.54	176 (13.63)	11497 (4.43)

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SUMMARY

- AI is a popular domain
- Machine learning is a subdiscipline of AI which gets a lot of attention
- We have seen some recent examples of novel developments and looked at the techniques behind it

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